# Supplementary Information: Non-Equilibrium Probes Reveal the Limits of Specialist and Foundation Machine-Learned Interatomic Potentials

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## Supplementary Information Roadmap

This Supplementary Information is organized to make the benchmark traceable from reference calculations to model evaluation. Section 1 describes the first-principles setup, NEB reference protocol, MACE training configurations, dataset provenance, and split controls. Section 2 provides the supporting analyses: RMSE benchmarking, structural and thermodynamic validation, molecular-dynamics property calculations, representation-learning descriptors, migration-path diagnostics, self-consistent MLIP-NEB tests, and SHAP surrogate interpretation. Together, these sections separate three observables that are otherwise easy to conflate: the DFT reference barrier on a converged path, the MLIP energy assigned to a fixed DFT path, and the barrier obtained after a model relaxes its own path.

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# 1 Detailed Computational Methods

## 1.1 First-Principles Reference Calculations

All first-principles calculations were performed using the QUANTUM ESPRESSO package [1]. The exchange–correlation energy was described using the Perdew–Burke–Ernzerhof (PBE) functional within the generalized gradient approximation (GGA) [2]. Long-range dispersion interactions, essential for accurately representing the interlayer physics of $Sb_2Te_3$, were included through the Grimme DFT-D3 correction scheme. Spin–orbit coupling (SOC) was incorporated through fully relativistic ONCV pseudopotentials (SG15 v1.2).

A plane-wave kinetic energy cutoff of 100 Ry and a charge density cutoff of 400 Ry were used for all calculations. Electronic occupations were treated with the cold-smearing scheme using a smearing width of 0.0147 Ry, which provides stable convergence for narrow-gap chalcogenides. Electronic states were iterated until the total electronic energy converged below $10^{-6}$ Ry, using a mixing parameter of 0.4 and a maximum of 200 electronic iterations per self-consistent cycle.

All structural relaxations were performed using the Broyden–Fletcher–Goldfarb–Shanno (BFGS) algorithm for both ionic and cell degrees of freedom, with force and pressure convergence thresholds of $10^{-3}$ Ry/Bohr and 0.5 kbar, respectively. A uniform Monkhorst–Pack mesh of $4 \times 4 \times 1$ was employed to sample the Brillouin zone for relaxations and Nudged Elastic Band (NEB) calculations. The $4 \times 2 \times 1$ supercell with dopant contained 81 atoms, corresponding to a two-quintuple-layer (2QL) slab of $Sb_2Te_3$ with one interstitial Cr atom, separated by sufficient vacuum to avoid spurious periodic interaction. Restart files were written periodically to ensure computational stability during long relaxations.

The training dataset was generated from ab initio molecular dynamics (AIMD) simulations performed in the NVT ensemble using a Langevin thermostat. These simulations covered a range of temperatures (300 K, 600 K, 1200 K) and Cr doping concentrations to ensure the model was exposed to a diverse set of thermodynamic and structural configurations. Each AIMD trajectory was run for 10 ps on a system of 120 atoms. To investigate the nature of atomic migration pathways, minimum energy paths (MEPs) and energy barriers were calculated using the Nudged Elastic Band (NEB) method. In the accompanying data release, each NEB path is reported with its task label, number of images, input structure provenance, restart lineage, convergence threshold, achieved maximum force, and activation barrier. Fixed-path scoring compares MLIP and QE energies on identical images, so the error it measures does not depend on whether the NEB band itself reached its force criterion. The within-gap reference path G1 (five images, PBE+D3) has a converged NEB with a forward barrier of 0.336 eV, and barrier errors in the main text are quantified on this path. For the other paths, the reference is the QE energy profile of the same image set (Table S6). Paths are named by class (G, within-gap; Q, gap-to-QL/into-QL; M, mixed; L and C, within-QL), and Table S7 gives the name, class, and classification basis of every path together with the internal identifier used in the data release. NEB paths are evaluation/reference data and were not merged into the FT-600K or FT-Multi$_T$ training splits. The released NEB-to-training exporters now fail closed when

real QE forces are unavailable, so a NEB geometry cannot silently acquire fabricated force labels.

## 1.2 MACE Model Training Protocols

### 1.2.1 Training Hyperparameters

All training and fine-tuning procedures were performed with consistent hyperparameters to ensure direct comparability. Unless otherwise stated, we used the Adam optimizer with an initial learning rate of $1 \times 10^{-3}$, a batch size of 32, a maximum of 1000 epochs, and early stopping based on validation-set force MAE. A 10% validation split was applied to all datasets.

Fine-tuning was initialized from the `MACE-MP-0 (PBE)` [3] foundation model. All model parameters were updated without freezing any layers, and all output heads were fine-tuned using the standard multi-head strategy. Stochastic Weight Averaging (SWA) was activated from epoch 100, and Exponential Moving Average (EMA) tracking ($\alpha = 0.99$) was applied throughout training. Energy and forces were read using the keys `energy` and `forces`, respectively, with force, energy, and stress loss weights of 100.0, 1.0, and 0.0. The training configuration employed $l_{\max} = 2$, cutoff radius $r_{\max} = 6.0$ Å. **[C8]** The reported fine-tuned models are documented as being trained on AIMD frames with real DFT energies and forces, not on NEB-derived exports lacking real force labels. The revised data manifest records source trajectory, split assignment, energy/force key mapping, and checksums for model-training files. The provenance audit covers 9,163 frames across the historical and grouped exports; the only nine exact zero-force frames are explicitly labelled one-atom Cr/Sb/Te isolated-atom $E_0$ references, with zero unclassified and zero NEB-derived zero-force frames. The affected NEB exporters fail closed when validated QE forces are unavailable; the complete manifest and audit code are in release commit `f5c6d19`.

### 1.2.2 Fine-Tuning Details

The multi-temperature fine-tuning model (FT-Multi_T) was constructed by combining configurations at 300 K, 600 K, and 1200 K. Each configuration was simulated at all three temperatures, ensuring that the Multi_T dataset is strictly balanced across temperatures without statistical or configurational bias. This design probes whether broader thermodynamic sampling improves model robustness and generalization.

Two data protocols are used. The frame-level baseline (Table S1) uses the original frame-level split, in which the FT-600K model was trained on approximately 5% of the 600 K AIMD pool (about 1000 configurations). The trajectory-grouped protocol (Table S2) trains the OMAT-FT-start FT-600K family on the full grouped training partition (1,786 configurations; validation/test partitions of 311/267 configurations). Scratch-5% is the low-data control: 89 configurations, 5% of the grouped training pool, drawn so that complete source trajectories stay intact. The 600 K data are representative of realistic thermoelectric operating conditions, lying near two-thirds of the melting point of $Sb_2Te_3$ and exhibiting significant anharmonic dynamics without structural degradation.

**[A2/C10]** The evaluation split assigns complete source trajectories to train, validation, or test. For the fine-tuning files used here, this gives 1,786/311/267 configurations for the full grouped split and 89/311/267 for Scratch-5%. Across 25 NEB images, no exact or near-duplicate training match is found: the minimum SOAP cosine distances are $5.4 \times 10^{-4}$ and $6.4 \times 10^{-4}$ (near-duplicate threshold $10^{-4}$), and the minimum comparable Cr-local RMSDs are 2.06 Å for both splits (threshold 0.05 Å). Grouped retraining results are reported below.

# 2 Supplementary Materials

## 2.1 MACE Model Performance Evaluation by RMSE

All developed machine-learned interatomic potentials (MLIPs) show low near-equilibrium energy and force errors in the current parity-plot audit (Figure S10), which compares model predictions to the reference DFT calculations. The performance of each model across the training, validation, and test sets is quantitatively summarized in Table S1. These values are the pointwise-accuracy baseline on the original frame-level split. Table S2 reports the trajectory-grouped, multi-seed retraining RMSE values and the within-gap barrier-error statistics.

**Table S1**: Performance metrics for MACE models across training, validation, and test sets. The FT-600K metrics are the mean ± standard deviation across three independent training runs with different random seeds.

| **Model** | **Train RMSE F** (meV/Å) | **Valid RMSE F** (meV/Å) | **Test RMSE F** (meV/Å) | **Test RMSE E** (meV/atom) |
|---|---|---|---|---|
| From Scratch | 67.1 | 76.1 | 75.2 | 1.0 |
| FT-600K | 20.7 ± 11.8 | 44.6 ± 2.9 | 37.2 ± 0.6 | 0.5 ± 0.0 |
| FT-MultiT | 20.3 | 49.1 | 45.5 | 0.5 |

Barrier statistics use the five DFT images of the within-gap path. The path registry (Table S7) assigns each path to the within-gap, gap-to-QL/into-QL, mixed, or within-QL class; each class is reported separately.

### 2.1.1 Task-conditioned distribution diagnostics

The grouped reference distribution was recomputed with a SOAP/PCA fit learned from the grouped training partition only and then applied without refitting to the validation, test, and fixed-path images. The grouped split contains 1,786 training, 311 validation, and 267 test configurations; the training partition contains 1,783 periodic AIMD frames and three isolated-atom calibration frames, while complete source trajectories remain assigned to one partition. The fixed-path images are displayed as a separate geometry class because the audited within-gap images contain 61 atoms, whereas the grouped AIMD cells contain 81–86 atoms. The resulting marginal and smooth two-dimensional densities are shown in the independently rendered reference

**Table S2**: **Trajectory-grouped comparison.** Values are means ± sample standard deviations across three independent training initializations unless noted. The within-gap column is the absolute fixed-path error relative to the 0.336 eV QE barrier of G1. FT-Multi$_T$ is evaluated on the frame-level baseline (Table S1) and in the MD and sliding tests.

| Model | Test $E$ RMSE (meV/atom) | Test $F$ RMSE (meV/Å) | Within-gap error (eV) |
|---|---|---|---|
| FT-600K (OMAT-FT start) | $40.7 \pm 5.0$ | $259.7 \pm 5.8$ | $0.47 \pm 0.12$ |
| Scratch | $30.5 \pm 3.6$ | $343.6 \pm 35.9$ | $1.20 \pm 0.27$ |
| Scratch-5% | $16.7 \pm 9.2$ | $341.5 \pm 15.0$ | $0.68 \pm 0.20$ |
| FT-600K (MPA-0 start; single initialization) | 38.7 | 278.5 | 0.24 |

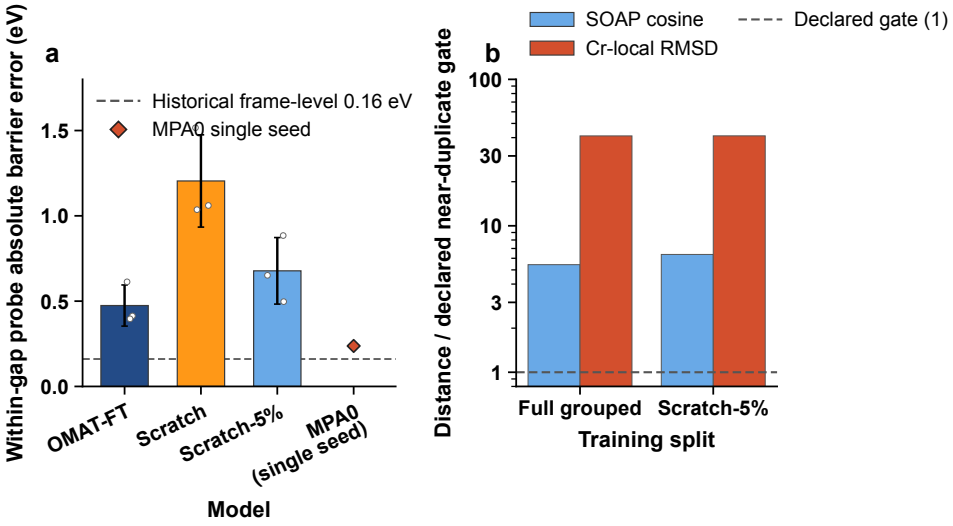



**Fig. S1**: **Grouped retraining and train–NEB overlap audit.** (a) Absolute fixed-path barrier errors for the controlled within-gap probe under the trajectory-grouped protocol. Open points are the three independent training initializations, bars/whiskers are the mean ± sample standard deviation, and the single-checkpoint MPA0 comparator is shown as a diamond. The dashed line marks the 0.16 eV single-run estimate obtained with the frame-level split, which the grouped FT-600K evaluation does not reproduce. (b) Minimum training-set SOAP and comparable Cr-local RMSD distances normalized by the declared near-duplicate gates. The dashed line at one is the declared near-duplicate gate; values above one lie outside that gate. RMSD is computed only for species-composition-compatible local environments, while SOAP covers all audited target images.

and MD panels in Fig. S2 and Fig. S4; the original Rockfish validation histories are provided separately in Fig. S3.

**Table S3**: **Stored MLIP-generated MD configuration-space audit.** The table reports every stored row available to the SOAP/PCA diagnostic. "Short-contact skip" denotes a row retained in the raw inventory but excluded from SOAP because the minimum interatomic distance was below 1.2 Å; it is a geometry-safety status, not a calibrated uncertainty estimate. The common MD velocity seed means that model/checkpoint branches are not independent velocity replicas.

| Family | Stored rows | SOAP-valid | Short-contact skip | PCA points |
|---|---|---|---|---|
| Foundation | 4,961 | 3 | 4,958 | 3 |
| Scratch | 4,001 | 1,310 | 2,691 | 1,310 |
| FT-600K | 4,601 | 4,601 | 0 | 4,601 |
| FT-Multi$_T$ | 4,001 | 4,001 | 0 | 4,001 |

The atlas also reports the available grouped-split static test RMSE evidence in the uniform manuscript order Foundation, Scratch, FT-600K, and FT-Multi$_T$. Foundation points are representative zero-shot grouped-test checkpoints, Scratch and FT-600K points are grouped multi-seed observations, and the FT-Multi$_T$ point is the frame-level evaluation. The MD clouds are configuration-space diagnostics, not DFT errors. Rows with an interatomic distance below 1.2 Å (4,958 of 4,961 Foundation rows and 2,691 of 4,001 Scratch rows) are marked as short-contact configurations and excluded from the SOAP projection.

The separate failure/CAR figure is an MLIP-only stage-resolved diagnostic based on the 639-candidate failure trajectories. It reports maximum force, paired model disagreement, stage-wise force envelopes, and distance to the grouped training representation. These panels describe failure progression and model disagreement. Source paths, hashes, and row counts are recorded in the accompanying distribution-atlas manifest.

### 2.1.2 Seed-aware kinetic uncertainty

**[A3]** Grouped retraining quantifies initialization uncertainty in the kinetic observable. For the within-gap path, the absolute fixed-path barrier errors are $0.47 \pm 0.12$ eV for FT-600K, $1.20 \pm 0.27$ eV for Scratch, and $0.68 \pm 0.20$ eV for Scratch-5%, across three independent training initializations (sample standard deviations across training runs). On the gap-to-QL/into-QL paths (Q1, Q2 and Q3), the Scratch errors are $0.51 \pm 0.11$, $0.72 \pm 0.55$ and $1.77 \pm 1.00$ eV, and the corresponding FT-600K errors are $9.60 \pm 0.57$, $10.40 \pm 0.44$ and $9.70 \pm 0.20$ eV. The first Scratch value is stable across initializations; the other two vary more between runs. Seed variation therefore does not account for the order-of-magnitude difference between FT-600K and Scratch on these sets.

A deeper analysis of the training dynamics reveals important characteristics of the models. The model trained from scratch exhibits a relatively small gap between

**Table S4**: **Class-resolved kinetic errors.** Entries are absolute fixed-path barrier errors relative to the QE barrier of each path (mean ± sample standard deviation over three independent training initializations); path names are defined in the SI path registry. G1 uses the 0.336 eV QE barrier; for the other paths the reference is the QE energy profile of the same images (SI NEB barrier table). Errors are energy-assignment errors on identical images, reported per path and not pooled.

| Pathway class | Model | Mean ± SD (eV) |
|---|---|---|
| Q1 (gap-to-QL/into-QL) | Scratch | $0.51 \pm 0.11$ |
| Q1 (gap-to-QL/into-QL) | FT-600K | $9.60 \pm 0.57$ |
| Q2 (gap-to-QL/into-QL) | Scratch | $0.72 \pm 0.55$ |
| Q2 (gap-to-QL/into-QL) | FT-600K | $10.40 \pm 0.44$ |
| G1 (within-gap) | Scratch | $1.20 \pm 0.27$ |
| G1 (within-gap) | FT-600K | $0.47 \pm 0.12$ |
| G1 (within-gap) | Scratch-5% | $0.68 \pm 0.20$ |
| Q3 (gap-to-QL/into-QL) | Scratch | $1.77 \pm 1.00$ |
| Q3 (gap-to-QL/into-QL) | FT-600K | $9.70 \pm 0.20$ |
| G2 (within-gap) | Scratch | $0.33 \pm 0.06$ |
| G2 (within-gap) | FT-600K | $0.21 \pm 0.18$ |

training (67.1 meV/Å) and validation (76.1 meV/Å) force RMSEs. While this suggests only moderate overfitting, its high error across all datasets indicates that it underfits the true physical interactions.

In contrast, the fine-tuned models display a more pronounced overfitting behavior, characterized by a large gap between their low training RMSEs and higher validation RMSEs. For instance, the FT-MultiT model shows a 142% increase in force RMSE from the training (20.3 meV/Å) to the validation set (49.1 meV/Å). This aggressive fitting is expected when fine-tuning on smaller, more specialized datasets. The high standard deviation in the FT-600K model's training RMSE (11.8 meV/Å) also suggests a sensitivity to initialization during the training process.

The fine-tuned models show lower errors than the scratch model on the current near-equilibrium test partition. Table S2 reports the grouped-split values used for the leakage-controlled comparison. RMSE is a pointwise baseline; the barrier and transport analyses below measure process fidelity.

## 2.2 Detailed Structural and Thermodynamic Analysis

The RDF analysis reveals that all MACE models, regardless of training strategy, successfully reproduce the key structural features of the Cr-doped $Sb_2Te_3$ system when compared to the AIMD ground truth (Fig. S11b).

All models correctly capture the primary coordination shells, with the first peak positions for Cr–Cr, Cr–Sb, and Cr–Te pairs occurring at approximately 3.0 Å, 3.2 Å,

and 2.8 Å, respectively, in excellent agreement with the AIMD reference. The peak heights and positions remain consistent across all training strategies (Figures S11c–f), indicating that the local atomic structure is well preserved regardless of whether the model was trained from scratch, used as a foundation model, or fine-tuned with temperature-specific data.

The primary observable difference between models lies in the smoothness of the RDF curves rather than their fundamental features. The scratch-trained model (Fig. S11d) exhibits slightly smoother RDF profiles compared to the fine-tuned variants (Figures S11e–f). This difference arises from practical computational considerations rather than fundamental accuracy: the scratch model, being more compact with fewer parameters, allowed for longer MD trajectories within the same computational budget, resulting in better statistical sampling. In contrast, the fine-tuned models, while maintaining larger parameter counts from their foundation architectures, required more computational resources per MD step, limiting the total simulation time and resulting in slightly noisier RDF profiles.

Notably, both fine-tuning strategies—whether using single-temperature (600 K) or multi-temperature AIMD data—yield RDF profiles that are nearly indistinguishable for this Cr-doped $Sb_2Te_3$ system. Within the scope of this test case, this similarity suggests that the structural descriptors learned by the models are not strongly sensitive to the temperature range of the fine-tuning dataset. While this observation is system-specific, it indicates that, for the structural properties examined here, the different training strategies converge to comparable representations of the local atomic environment. The consistent reproduction of key RDF features across all models supports the conclusion that each MLIP captures the essential bonding motifs relevant to this particular material.

## 2.3 Thermodynamic Ensemble Stability

A detailed analysis of the pressure profiles reveals subtle but important differences between models. The zero-shot MACE Foundation model equilibrates to a slightly different average pressure than the models trained or fine-tuned on our in-house AIMD data. This is likely a consequence of the foundation model being trained on a vast dataset of materials at their 0K equilibrium volumes. A minor mismatch between the foundation model's predicted equilibrium lattice parameters and the true DFT values for our specific Cr:$Sb_2Te_3$ system at 600K manifests as a persistent non-zero average pressure in an NVT simulation. The fine-tuned and scratch models, having been trained explicitly on data from this system's true ensemble, do not exhibit this deviation.

## 2.4 Transport Property Analysis

**[C9]** The models show similar RDF and VACF behavior but different mobility. Under the common 200 ps protocol (6 blocks), the Foundation and Scratch MSDs plateau (fit $R^2 \leq 0.01$), so $D$ is indistinguishable from zero, whereas the FT-MultiT MSD grows approximately linearly ($R^2 = 0.86$) with $D = (4.8 \pm 4.8) \times 10^{-8}$ $\mathrm{cm^2\,s^{-1}}$. The Green–Kubo thermal conductivities are at most $2 \times 10^{-2}$ $\mathrm{W\,m^{-1}\,K^{-1}}$ for all models

and within block uncertainty of zero. The FT-600K MD run used a shorter trajectory (2,000 steps, output every 5 steps) and is shown for reference only.

**Table S5**: **Molecular-dynamics transport metrics.** $D$ and $\kappa$ with block standard deviations (6 blocks each). †The FT-600K run used a shorter trajectory with a different output interval and is shown for reference; the other three models share the same protocol.

| Model | $D$ ($cm^2/s$) | $R^2$ (MSD fit) | $\kappa$ ($W\,m^{-1}\,K^{-1}$) | Protocol |
|---|---|---|---|---|
| Foundation | $(0.7 \pm 7.9) \times 10^{-9}$ | 0.01 | $(2.4 \pm 8.6) \times 10^{-3}$ | common |
| Scratch | $(-3.4 \pm 3.2) \times 10^{-9}$ | 0.00 | $(0.9 \pm 1.3) \times 10^{-2}$ | common |
| FT-600K† | $(3.9 \pm 4.0) \times 10^{-8}$ | 0.90 | $(0.9 \pm 1.2) \times 10^{-2}$ | shorter run |
| FT-MultiT | $(4.8 \pm 4.8) \times 10^{-8}$ | 0.86 | $(2.0 \pm 1.7) \times 10^{-2}$ | common |

†The FT-600K 600 K run used 2,000 steps with output every 5 steps, against 100,000 steps with output every 200 steps for the other models, and is shown for reference.

$D$: block-averaged MSD slope (6 blocks, fit window 5–200 ps). $\kappa$: Green–Kubo running integral averaged over 100–200 ps (6 blocks).

## 2.5 Molecular Dynamics Simulations

### 2.5.1 Simulation Protocol

MD simulations were performed using the Atomic Simulation Environment (ASE) with the following protocol:

- **Integrator**: Langevin dynamics for NVT ensemble or Nosé-Hoover for NPT ensemble
- **Timestep**: 1.0 fs
- **Friction coefficient**: $\gamma = 0.01$ $fs^{-1}$ (Langevin)
- **Barostat parameters**: $\tau_p = 1000$ fs, $P = 1$ bar (NPT only)
- **Equilibration**: 50,000 steps (50 ps)
- **Production**: 200,000 steps (200 ps) for property calculations
- **Sampling interval**: Every 100 steps for analysis
- **System size**: 20×20×1 supercell (4000 atoms)

Simulations were conducted at two temperatures: 300 K and 600 K, to assess model performance under different thermal conditions. Initial velocities were assigned according to the Maxwell-Boltzmann distribution with removal of center-of-mass motion and angular momentum.

## 2.6 Property Calculations

### 2.6.1 Structural Properties

The radial distribution function (RDF) $g(r)$ was computed for all unique atom pairs:

$$g_{\alpha\beta}(r) = \frac{V}{4\pi r^2 \Delta r N_\alpha N_\beta} \left\langle \sum_{i \in \alpha} \sum_{j \in \beta} \delta(r - r_{ij}) \right\rangle \tag{1}$$

where $\alpha$ and $\beta$ denote atomic species, $V$ is the system volume, $N_\alpha$ is the number of atoms of species $\alpha$, and the angle brackets denote time averaging.

### 2.6.2 Dynamic Properties

The mean square displacement (MSD) was calculated for each atomic species:

$$\mathrm{MSD}_\alpha(t) = \left\langle \frac{1}{N_\alpha} \sum_{i \in \alpha} |\mathbf{r}_i(t) - \mathbf{r}_i(0)|^2 \right\rangle \tag{2}$$

Diffusion coefficients were extracted from the linear regime of MSD using the Einstein relation:

$$D_\alpha = \lim_{t \to \infty} \frac{\mathrm{MSD}_\alpha(t)}{6t} \tag{3}$$

The velocity autocorrelation function (VACF) was computed as:

$$C_v(t) = \frac{\langle \mathbf{v}(t) \cdot \mathbf{v}(0) \rangle}{\langle \mathbf{v}(0) \cdot \mathbf{v}(0) \rangle} \tag{4}$$

### 2.6.3 Thermodynamic Properties

Average potential energy per atom, temperature, and volume (for NPT simulations) were calculated over the production phase:

$$\langle E \rangle = \frac{1}{N_{\mathrm{frames}}} \sum_{i=1}^{N_{\mathrm{frames}}} \frac{E_{\mathrm{pot}}^{(i)}}{N_{\mathrm{atoms}}} \tag{5}$$

with corresponding standard deviations to assess thermal fluctuations.

### 2.6.4 Transport Properties

For thermal conductivity calculations, the heat flux vector was computed:

$$\mathbf{J} = \frac{1}{V} \left[ \sum_i e_i \mathbf{v}_i + \frac{1}{2} \sum_{i \neq j} (\mathbf{F}_{ij} \cdot \mathbf{v}_j) \mathbf{r}_{ij} \right] \tag{6}$$

where $e_i$ is the per-atom energy, $\mathbf{v}_i$ is the velocity, $\mathbf{F}_{ij}$ is the force between atoms $i$ and $j$, and $\mathbf{r}_{ij}$ is their separation vector. The heat flux autocorrelation function (HFACF) was then computed for subsequent Green-Kubo analysis.

## 2.7 Representation Learning Feature Extraction

The following physically interpretable descriptors were extracted from MD trajectories:

1. **Energy landscape**: Total potential energy per atom
2. **Force fields**: 3N-dimensional force vectors for all atoms
3. **Structural order parameters**: Radial distribution histograms computed with 100 bins up to 5.0 Å cutoff
4. **SOAP descriptors**: Smooth Overlap of Atomic Positions with:
   - Cutoff radius: 5.0 Å
   - Number of radial basis functions: 8
   - Maximum angular momentum: 4
   - Gaussian width: 0.3 Å

[C12] The feature matrix excludes the earlier always-zero force-sensitivity placeholder; it carried no physical information and is not counted in the reported descriptor dimension. The sensitivity analysis evaluates $t$-SNE perplexities 5, 30, and 50 and nominal PHATE seeds 0–2.

## 2.8 Evaluation Metrics Definitions

- **Force MAE**: $\frac{1}{3N}\sum_{i=1}^{N}\sum_{\alpha=x,y,z}|F_{i,\alpha}^{\mathrm{MLIP}} - F_{i,\alpha}^{\mathrm{DFT}}|$
- **Energy MAE**: $\frac{1}{N}|E^{\mathrm{MLIP}} - E^{\mathrm{DFT}}|$
- **RMSD Growth Rate**: Linear fit slope of $\mathrm{RMSD}(t) = \sqrt{\frac{1}{N}\sum_{i=1}^{N}|\mathbf{r}_i(t) - \mathbf{r}_i(0)|^2}$
- **Silhouette Score**: Average of $\frac{b_i - a_i}{\max(a_i, b_i)}$ where $a_i$ is mean intra-cluster distance and $b_i$ is mean nearest-cluster distance

## 2.9 Detailed Migration Pathway Analysis

Table S8 reports the three-initialization absolute fixed-path errors for FT-600K, Scratch-5%, and Scratch on the 61-atom paths. Within-gap paths (G1, G2) and gap-to-QL/into-QL paths (Q1, Q2, Q3) are listed separately because they probe different lattice distortions.

The 61-atom fixed-image evaluations are single-point model energies on the DFT images of each path. Here "gap-to-QL/into-QL" denotes motion from the vdW gap toward or into the interior of a covalently bonded QL, and "within-gap" denotes motion that stays in the vdW gap. Path tortuosity describes the winding geometry of a trajectory, whereas lattice distortion describes the displacement of the surrounding QL; the within-QL label is based on the latter.

### 2.9.1 In-Gap Diffusion

[A5] The Scratch-5% control is a random-initialization low-data comparison with 89 grouped training configurations and 311/267 validation/test configurations. Its within-gap fixed-path error is $0.68 \pm 0.20$ eV, versus $0.47 \pm 0.12$ eV for FT-600K and $1.20 \pm 0.27$ eV for Scratch, both trained on 1,786 configurations. The from-scratch

**Table S6**: **QE NEB barriers of the Cr migration paths.** Forward and reverse activation energies relative to the first image of each path (path names in Table S7); the status column records whether the QE run reached its force criterion.

| Name | Class | $N_{\mathrm{img}}$ | $E_{\mathrm{act}}^{\rightarrow}$ (eV) | $E_{\mathrm{act}}^{\leftarrow}$ (eV) | NEB status |
|---|---|---|---|---|---|
| G1 | within-gap | 5 | 0.34 | 0.02 | converged |
| G2 | within-gap | 5 | 0.58 | 0.02 | not converged |
| Q1 | gap-to-QL/into-QL | 5 | 2.79 | 1.02 | not converged |
| Q2 | gap-to-QL/into-QL | 5 | 4.00 | 3.95 | not converged |
| Q3 | gap-to-QL/into-QL | 5 | 1.10 | 0.75 | not converged |
| M1 | mixed | 5 | 1.03 | 0.21 | not converged |
| L1 | within-QL | 5 | 1.76 | 0.99 | converged |
| C1 | within-QL | 10 | 1.03 | 1.03 | converged |
| C2 | within-QL | 10 | 1.56 | 2.11 | converged |
| C3 | within-QL | 10 | 1.69 | 1.67 | converged |
| C4 | within-QL | 10 | 2.06 | 1.54 | converged |
| C5 | within-QL | 10 | 1.41 | 1.07 | converged |

error is not monotonic in training-set size on this path, and the fine-tuned model has the lowest error.

For a matched foundation-model comparison, we evaluated ten single-checkpoint models on the same five fixed images. The four main representatives (MP-0b3, MPA0, OMAT-0 medium, and MH-1) produced barriers of 1.50, 1.79, 1.60, and 1.84 eV, respectively, against the DFT reference of 0.34 eV. The ten individual profiles are shown in Fig. S14a and the family means with pointwise $\pm 1$ standard-deviation bands are shown in Fig. S14b. The family bands describe variation among sub-models within each foundation family, not training-initialization uncertainty. The profiles are smooth but systematically elevated relative to the DFT reference, so equilibrium accuracy and fixed-path migration fidelity are distinct observables.

### 2.9.2 Higher-Distortion and Within-QL Paths

Within-QL migration is defined here as Cr entering the interior of a covalently bonded QL, which can require substantial lattice distortion. The path registry (Table S7) assigns each path to the within-gap, gap-to-QL/into-QL, mixed, or within-QL class, and errors are reported per path because the classes probe different distortions.

On the gap-to-QL/into-QL paths the FT-600K energies deviate from the QE energies by 9.6–10.4 eV in every training initialization (Table S8), against QE barriers of 1.1–4.0 eV, so the fine-tuned model assigns barriers above 10 eV to these paths. The from-scratch models deviate by 0.3–1.8 eV with larger run-to-run spread. Grouped multi-seed retraining gives the model-level uncertainty for these comparisons.

### 2.9.3 Implications for MLIP Development

These findings highlight critical considerations for practical MLIP deployment:

**Table S7**: **Cr migration path registry.** Each path is named by class and index. The audit ID is the identifier used in the released data and scripts. Classes follow the geometry label of the path definition (path queue) or, for M1, L1 and C1–C5, the exact image coordinates (see the classification basis); model errors are reported per path and are not pooled across classes.

| Name | Class | Atoms | Audit ID | Classification basis |
|---|---|---|---|---|
| G1 | within-gap | 61 | `1-4` | Cr stays in the vdW gap between sites defined as in-gap in the path queue; barrier reference path |
| G2 | within-gap | 61 | `1-7` | Cr stays in the vdW gap between sites defined as in-gap in the path queue |
| Q1 | gap-to-QL/into-QL | 61 | `1-2` | Path queue sites place Cr moving from the vdW gap into the quintuple layer |
| Q2 | gap-to-QL/into-QL | 61 | `1-3` | Path queue sites place Cr moving from the vdW gap into the quintuple layer |
| Q3 | gap-to-QL/into-QL | 61 | `1-5` | Path queue sites place Cr moving from the vdW gap into the quintuple layer; the queue asks for a geometry recheck |
| M1 | mixed | 61 | `1-6` | Exact image coordinates give a mixed gap/QL transition |
| L1 | within-QL | 61 | `1-8` | Exact image coordinates give probable deep penetration |
| C1 | within-QL | 81 | `neb_1` | Continuous-geometry penetration score 0.69–0.74 (gap score 0.21–0.35); five-path chain |
| C2 | within-QL | 81 | `neb_2` | Continuous-geometry penetration score 0.69–0.74 (gap score 0.21–0.35); five-path chain |
| C3 | within-QL | 81 | `neb_3` | Continuous-geometry penetration score 0.69–0.74 (gap score 0.21–0.35); five-path chain |
| C4 | within-QL | 81 | `neb_4` | Continuous-geometry penetration score 0.69–0.74 (gap score 0.21–0.35); five-path chain |
| C5 | within-QL | 81 | `neb_5` | Continuous-geometry penetration score 0.69–0.74 (gap score 0.21–0.35); five-path chain |

**Table S8**: **Path-resolved fixed-path errors for the grouped models.** Absolute errors (eV), mean ± sample standard deviation across three independent training initializations for FT-600K, Scratch-5%, and full Scratch. The QE column gives the forward barrier of the QE image set of each path. Errors are energy-assignment errors on identical images, reported per path and not pooled across classes. Path names are defined in Table S7.

| Path | Class | QE barrier | FT-600K | Scratch-5% | Scratch |
|---|---|---|---|---|---|
| G1 | within-gap (reference path) | 0.34 | $0.47 \pm 0.12$ | $0.68 \pm 0.20$ | $1.20 \pm 0.27$ |
| G2 | within-gap | 0.58 | $0.21 \pm 0.18$ | $0.25 \pm 0.21$ | $0.33 \pm 0.06$ |
| Q1 | gap-to-QL/into-QL | 2.79 | $9.60 \pm 0.57$ | $0.33 \pm 0.31$ | $0.51 \pm 0.11$ |
| Q2 | gap-to-QL/into-QL | 4.00 | $10.40 \pm 0.44$ | $1.22 \pm 0.66$ | $0.72 \pm 0.55$ |
| Q3 | gap-to-QL/into-QL | 1.10 | $9.70 \pm 0.20$ | $1.07 \pm 0.31$ | $1.77 \pm 1.00$ |

- **Validation Insufficiency**: Testing only on stable configurations masks critical failures at transition states
- **Hidden Extrapolation**: Models can appear accurate at trajectory endpoints while failing at crucial transition states
- **Data Quality over Quantity**: Task-specific training data outweighs larger but less relevant datasets
- **Foundation Model Limitations**: Pre-trained models require careful adaptation for processes involving significant atomic rearrangement

## 2.10 Nudged Elastic Band (NEB) Benchmark for Cr Migration

We assess each model on a Cr migration event with two operators (Fig. S15). Fixed-path scoring (panel a) assigns a barrier to the five DFT images of the 61-atom, one-Cr within-gap path; the QE barrier is 0.336 eV and the models assign 1.02 (Foundation), 4.54 (Scratch), 0.50 (FT-600K) and 0.82 eV (FT-MultiT). Self-relaxed NEB (panel b) lets the model move the images under its own forces; this is a stability test of model-in-the-loop path generation, and it is run on the 82-atom, two-Cr cell.

The Scratch band does not converge: the maximum image energy rises to $10^2$–$10^4$ eV above the endpoint within about 100 optimizer steps and stays there for the remaining 1,300 steps. The Scratch and fine-tuned models (FT-600K and FT-MultiT) all show a rapid increase in the maximum image force ($f_{max}$) during NEB optimization, and the runs terminate before a converged band is obtained. Self-relaxed NEB therefore detects a force-feedback instability that the fixed-path operator, which never moves the images, cannot.

**[A1]** The fixed-path and self-relaxed NEB evaluations are different measurement operators. In the fixed-path evaluation, the DFT-relaxed images of the 61-atom, one-Cr within-gap path are held fixed; the QE barrier is 0.336 eV (0.34 eV) and the Foundation model assigns 1.02 eV (+0.69 eV). The 0.41 eV Foundation value that SI Fig. S3 of the previous version showed belonged to a different path (82 atoms, two Cr, twelve images) and has been removed. Barrier ranking uses the fixed-path operator; self-relaxed NEB is used as a stability test.

## 2.11 Representation and Surrogate Diagnostics

[**A6/C11**] The representation analysis uses 404 trajectory frames and 6,191 descriptors after removing the all-zero force-sensitivity placeholder. In the original descriptor space, the overall Euclidean silhouette is 0.33 and the cosine silhouette is 0.28. After z-scoring each descriptor, the Euclidean silhouette is $-0.02$; a 95% variance PCA of the z-scored matrix gives $-0.03$ in 364 dimensions. Because the descriptor vector combines total energy, forces, RDF features, and SOAP components, the raw-space value separates the models' recorded outputs largely through the scale of these components, and the z-scored value shows no separation once scale is removed.

[**A6/C11**] The projection audit gives a $t$-SNE silhouette of $0.40 \pm 0.03$ across perplexities 5, 30, and 50, with a range of 0.36–0.42 and trustworthiness of 0.963–0.973. PHATE gives $0.47 \pm 0.00$ across three nominal seeds. At fixed perplexity, the nominal $t$-SNE seeds are bitwise-identical under the fixed PCA initialization; the reported spread therefore reflects perplexity sensitivity and is not treated as independent seed uncertainty. The projections are qualitative diagnostics consistent with different descriptor distributions.

[**A6/C13**] For the SHAP analysis, the corrected frame-aligned workflow uses 150 configurations, 390 SOAP descriptors, and 17 structural descriptors. Over 30 random seeds, the 5-fold cross-validated surrogate $R^2$ values (mean $\pm$ s.d.) are $0.973 \pm 0.008$ for FT-600K, $0.858 \pm 0.051$ for FT-MultiT, and $0.970 \pm 0.013$ for Scratch, with corresponding MAEs of $0.045 \pm 0.005$, $0.027 \pm 0.002$, and $0.027 \pm 0.004$ eV. The most frequent top-ranked descriptor is `Cr-Sb_n43_l0` for FT-MultiT (30/30 seeds), `Cr-Sb_n13_l0` for FT-600K (20/30), and `Cr-Te_n32_l2` for Scratch (19/30). The largest FT-MultiT feature is `Cr-Sb_n43_l0`; Cr–Cr descriptors occur in some rankings but are not a stable dominant channel across models. These values describe the surrogate's sensitivity to predictors of absolute energy error and do not establish a causal attribution within MACE.

[**A6**] A complementary endpoint campaign contains 846 fixed-cell native-MACE FIRE relaxations across six Cr/V/Mn–$Sb_2Te_3$/$Bi_2Te_3$ systems and three perturbation arms; all tasks pass the finite-value, topology, minimum-distance, displacement, and convergence gates. It measures the local stability of native-MACE endpoint relaxations under dopant displacement (endpoint appendix); it perturbs the dopant position and does not isolate a single SOAP channel.

# 3 Six-system multi-arm endpoint perturbation campaign

We evaluated the response of six dopant–host systems, Cr/V/Mn embedded in $Sb_2Te_3$ and $Bi_2Te_3$, to three endpoint-relaxation arms: the historical unperturbed endpoint (Base), a +0.20 Å dopant displacement (Perturb A), and a displacement of $(-0.10, +0.14, +0.10)$ Å (Perturb B). The campaign contains 47 endpoint sites per system and arm, for 846 formal MACE relaxation tasks. All tasks passed the finite-value, topology, minimum-distance, displacement, and scientific-success gates. These results are model-relaxation observables from Foundation-MACE with fixed-cell ASE FIRE relaxation; they are not DFT, magnetic, or oxidation-state assignments.

**Table S9**: **Separation metrics across projected and descriptor spaces.** Projected values are baselines. The raw descriptor-space value depends on the relative scale of the energy, force, RDF, and SOAP components; the z-scored and PCA values remove that dependence.

| Space | Overall silhouette | Interpretation |
|---|---|---|
| Original 6,191-d, Euclidean | 0.33 | Raw descriptor space |
| Original 6,191-d, cosine | 0.28 | Cosine metric |
| Original 6,191-d, $z$-scored Euclidean | $-0.02$ | Scale removed |
| PCA 95%, $z$-scored Euclidean | $-0.03$ | 364 dimensions |
| $t$-SNE, $p \in \{5, 30, 50\}$, 3 nominal seeds | $0.40 \pm 0.03$ | Range 0.36–0.42; trustworthiness 0.963–0.973 |
| PHATE 2D, 3 nominal seeds | $0.47 \pm 0.00$ | Range 0.47265–0.47277 |

**Table S10**: **SHAP surrogate fidelity.** 5-fold cross-validated $R^2$ and MAE (mean $\pm$ s.d. over 30 seeds) of the gradient-boosting surrogate for the absolute energy error, on the 150-frame, frame-aligned workflow.

| Model | $n$ | 5-fold CV $R^2$ | 5-fold CV MAE (eV) |
|---|---|---|---|
| FT-600K | 150 | $0.973 \pm 0.008$ | $0.045 \pm 0.005$ |
| FT-MultiT | 150 | $0.858 \pm 0.051$ | $0.027 \pm 0.002$ |
| Scratch | 150 | $0.970 \pm 0.013$ | $0.027 \pm 0.004$ |

Mean $\pm$ s.d. over 30 random seeds (surrogate and CV-fold seeds).

## 3.1 Interpretation and reproducibility

The complete endpoint-level data used for these figures are provided in `revision1/appendix/data/endpoint_level_summary.csv` and its JSON counterpart. The summary table is generated from `revision1/appendix/data/multiarms_summary.csv`. The plotting source is preserved as `revision1/appendix/scripts/plot_campaign_atlas.py`; it uses the project Plot Atlas primitives and regenerates the three PDF, PNG, and SVG figures from the supplied data. Overleaf compiles the checked-in PDF assets; the data and script are included alongside them so that the figures remain auditable and reproducible outside the hosted compiler.

The dominant qualitative result is that all three arms are scientifically accepted, while perturbation magnitude and response depend strongly on both host and dopant. In particular, the $Bi_2Te_3$ systems exhibit larger endpoint displacements than the corresponding $Sb_2Te_3$ systems, whereas the final force distributions remain tightly concentrated near the FIRE acceptance threshold. These patterns support using endpoint perturbations as a controlled local-stability probe, but do not establish transferability to DFT energetics or to finite-temperature migration barriers.

**Table S11**: **Measured multi-arm endpoint-relaxation summary.** Values are medians over the 47 historical endpoint sites for each of six host–dopant systems and each perturbation arm. "Conv." reports the number of FIRE relaxations reaching $f_{\max} \leq 0.03$ eV Å$^{-1}$; $E_f$ is the final MACE energy, $F_{\max}$ the final maximum force, $d_{\max}$ the maximum atomic displacement from the perturbed input, and "steps" the FIRE step count. All 846 formal tasks passed the finite-value, topology, minimum-distance, displacement, and scientific-success gates. The Foundation-MACE model and fixed-cell FIRE protocol are the same across arms; this table reports model-relaxation observables, not DFT or magnetic conclusions.

| | **Base** | | | | **Perturb A:** $+x\,0.20$ | | | | **Perturb B:** $(-x, +y, +z)$ | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **System** | **Conv.** (47) | $E_f$ (eV) | $F_{\max}$ (eV/Å) | $d_{\max}$ (Å) | **Conv.** (47) | $E_f$ (eV) | $F_{\max}$ (eV/Å) | $d_{\max}$ (Å) | **Conv.** (47) | $E_f$ (eV) | $F_{\max}$ (eV/Å) | $d_{\max}$ (Å) |
| Cr@Sb2Te3 | 47/47 | -299.68 | 0.03 | 0.02 | 47/47 | -299.69 | 0.03 | 0.20 | 47/47 | -299.70 | 0.03 | 0.19 |
| V@Sb2Te3 | 47/47 | -299.47 | 0.03 | 0.13 | 47/47 | -299.48 | 0.03 | 0.23 | 47/47 | -299.46 | 0.03 | 0.24 |
| Mn@Sb2Te3 | 47/47 | -299.34 | 0.03 | 0.13 | 47/47 | -299.34 | 0.03 | 0.23 | 47/47 | -299.34 | 0.03 | 0.23 |
| Cr@Bi2Te3 | 47/47 | -300.84 | 0.03 | 0.47 | 47/47 | -300.82 | 0.03 | 0.49 | 47/47 | -300.83 | 0.03 | 0.47 |
| V@Bi2Te3 | 47/47 | -300.59 | 0.03 | 0.50 | 47/47 | -300.61 | 0.03 | 0.53 | 47/47 | -300.58 | 0.03 | 0.55 |
| Mn@Bi2Te3 | 47/47 | -300.71 | 0.03 | 0.50 | 47/47 | -300.71 | 0.03 | 0.53 | 47/47 | -300.71 | 0.03 | 0.53 |

*Arm definitions.* Base is the unperturbed historical endpoint. Perturb A displaces the dopant by $(+0.20, 0, 0)$ Å; Perturb B uses $(-0.10, +0.14, +0.10)$ Å(norm 0.20 Å). Medians summarize distributions and should not be read as endpoint-level equivalence.

# References


[1] Giannozzi, P., Baroni, S., Bonini, N., Calandra, M., Car, R., Cavazzoni, C., Ceresoli, D., Chiarotti, G.L., Cococcioni, M., Dabo, I., *et al.*: Quantum espresso: a modular and open-source software project for quantumsimulations of materials. Journal of physics: Condensed matter **21**(39), 395502 (2009)

[2] Perdew, J.P., Burke, K., Ernzerhof, M.: Generalized gradient approximation made simple. Physical review letters **77**(18), 3865 (1996)

[3] Batatia, I., Kovacs, D.P., Simm, G., Ortner, C., Csányi, G.: Mace: Higher order equivariant message passing neural networks for fast and accurate force fields. Advances in neural information processing systems **35**, 11423–11436 (2022)

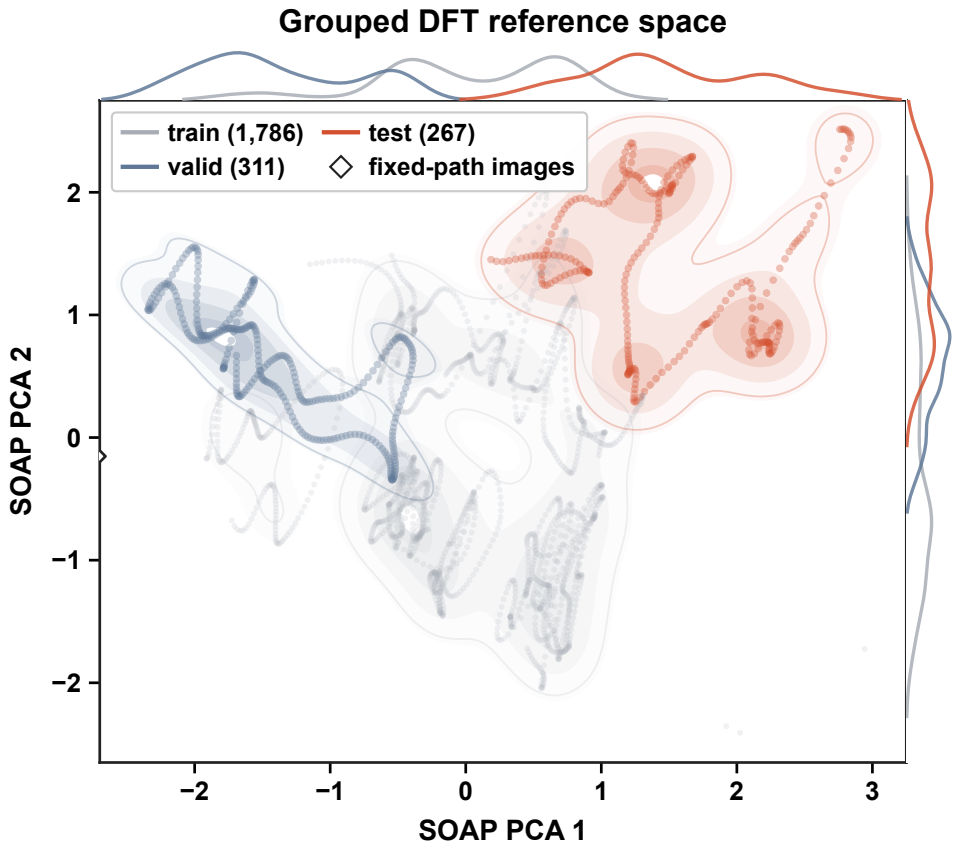



**Fig. S2**: **Grouped DFT reference space.** SOAP/PCA is fitted on the trajectory-grouped training partition only and applied without refitting to 1,786 training, 311 validation, 267 test, and five fixed-path images of the within-gap path G1. The training partition contains 1,783 periodic AIMD frames and three isolated-atom calibration frames; complete source AIMD trajectories are assigned to one partition. Soft two-dimensional density surfaces and calibrated marginal supports show the grouped split, while diamonds identify the fixed-path images. The latter are kept as a separate geometry class because the audited G1 images contain 61 atoms, whereas the grouped AIMD cells contain 81–86 atoms.

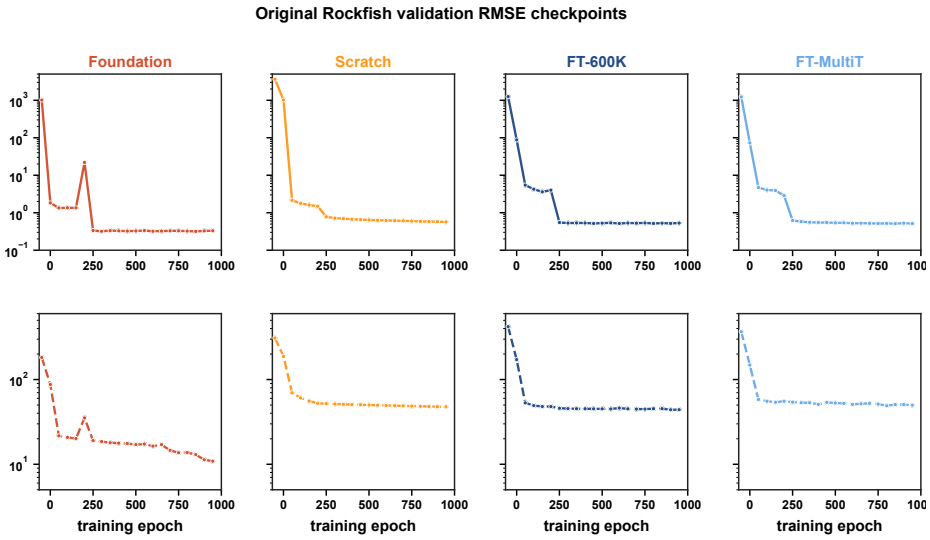



**Fig. S3**: **Validation RMSE during training.** The four columns follow the order Foundation, Scratch, FT-600K, and FT-MultiT. The upper row reports energy RMSE per atom and the lower row reports force RMSE, both converted from the original JSONL training logs to meV-based units. Each curve contains the unique default-head `mode=eval` checkpoints from the corresponding run 123 log; duplicate log checkpoints are retained in the raw table but omitted from the plotted trace. The curves are single-run validation histories from the original training campaigns. Source paths and SHA256 values are recorded in the accompanying RMSE manifest.

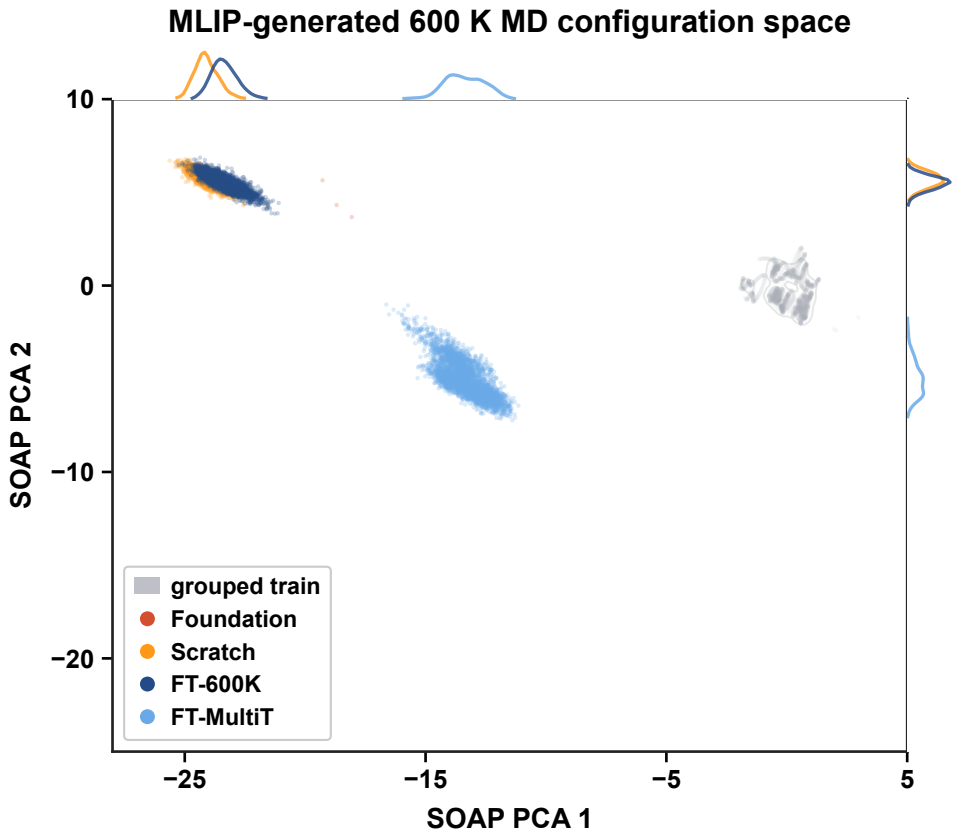



**Fig. S4**: **MLIP-generated 600 K MD configuration space.** SOAP/PCA distributions from the stored MLIP-generated continuation segments are overlaid on the grouped-training density using the same PCA transform as panel (a). The stored rows are Foundation 4,961 (3 SOAP-valid and 4,958 short-contact skips), Scratch 4,001 (1,310 valid and 2,691 short-contact skips), FT-600K 4,601 valid, and FT-MultiT 4,001 valid. Short-contact rows (minimum interatomic distance below 1.2 Å) are excluded from SOAP. The MD branches share a common velocity seed.

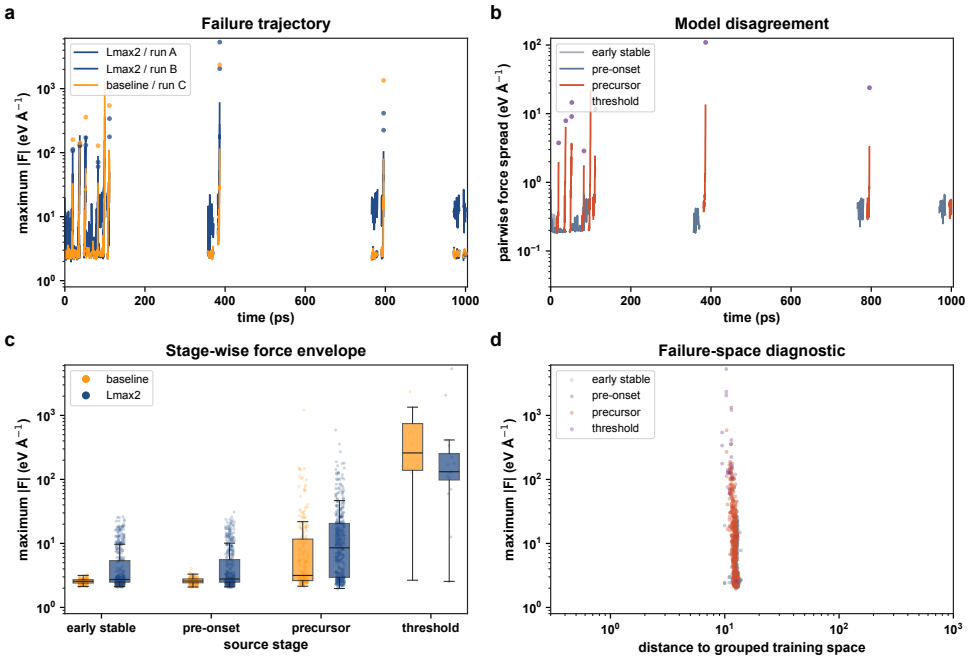



**Fig. S5**: **Stage-resolved failure and model-disagreement diagnostic.** (a) Maximum force along the 639-candidate failure trajectories for the baseline and $l_{\max} = 2$ branches. (b) Paired model force disagreement across the same stage windows. (c) Stage-wise force envelopes for the two model branches. (d) Maximum force as a function of distance to the grouped training representation. The stage-balanced samples describe failure progression and model disagreement.

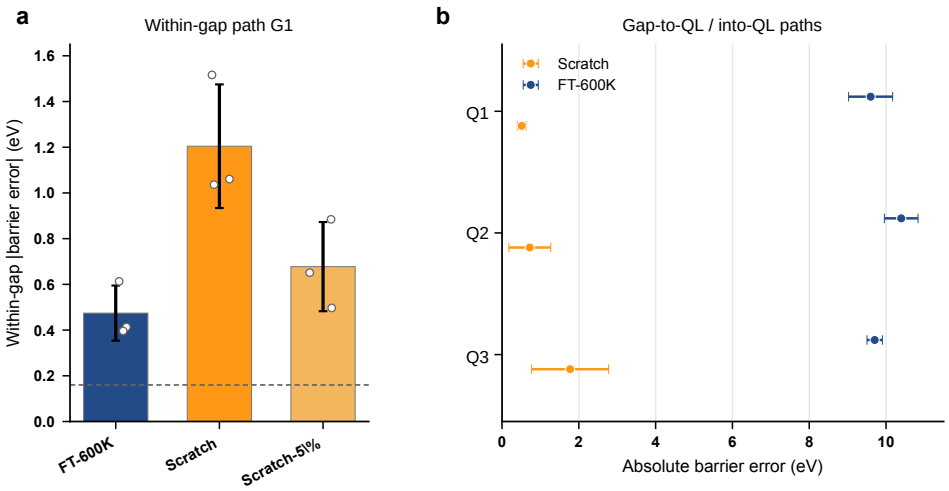
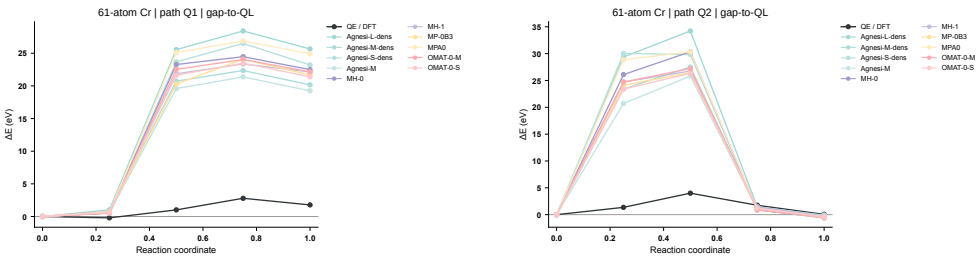



**Fig. S6**: **Seed-resolved kinetic uncertainty.** The grouped within-gap comparison and representative gap-to-QL/into-QL errors are shown with the individual-run observations and mean ± sample standard deviation. For Scratch, Q1 is stable across initializations and Q2 and Q3 vary more between runs.




(a) Q1. (b) Q2.

**Fig. S7**: **Fixed-path energy profiles for gap-to-QL/into-QL paths.** Model energies on the DFT images of Q1 and Q2, with the QE profile for comparison. Seed-resolved errors for these paths are given in Table S8.

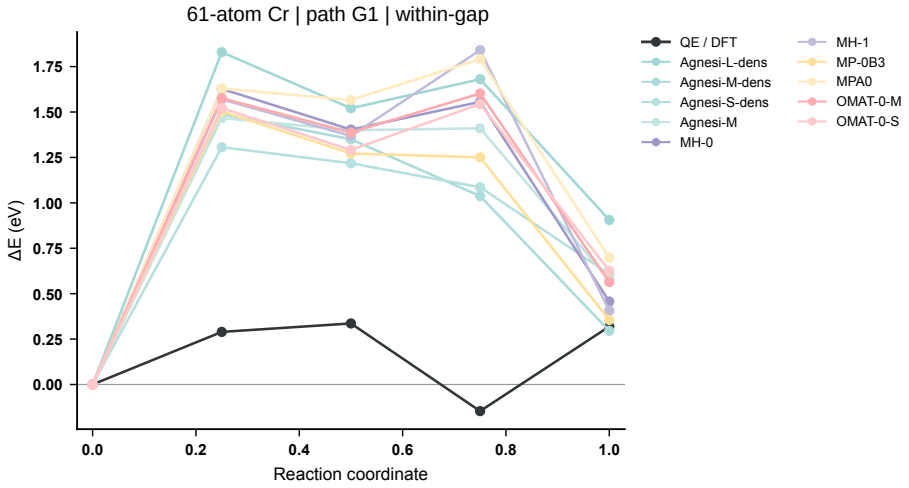
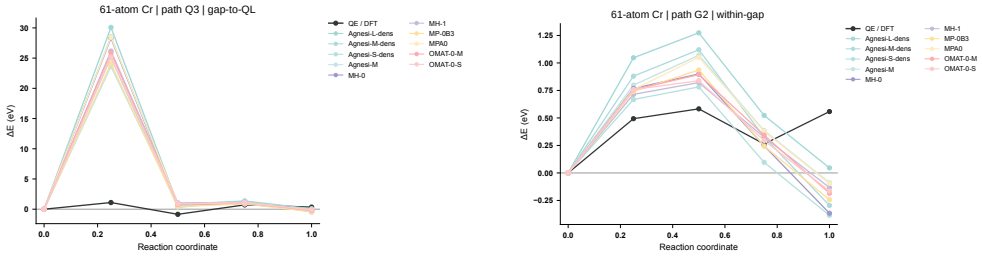



**Fig. S8**: **Fixed-path energy profile for the within-gap path G1.** Model energies on the common DFT images used for the barrier comparison. The grouped three-initialization errors are given in Table S2 and Fig. S6.




(a) Q3 (gap-to-QL/into-QL). (b) G2 (within-gap).

**Fig. S9**: **Fixed-path energy profiles for Q3 and G2.** Q3 is a gap-to-QL/into-QL path and G2 a within-gap path; each is reported separately.

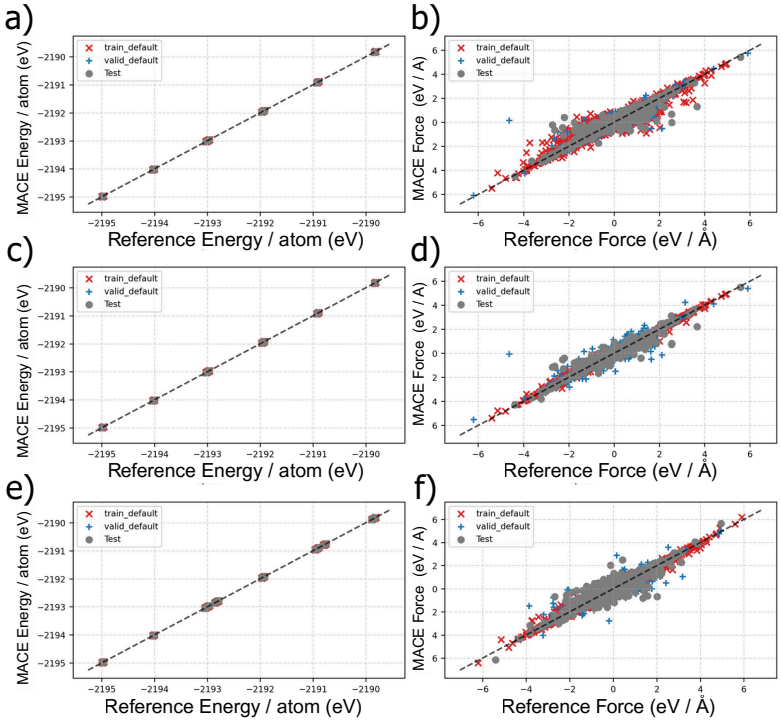



**Fig. S10**: **Parity plots comparing MACE-predicted energies and forces against DFT reference calculations for the test set.** (a, b) Model trained from scratch, (c, d) fine-tuned 600 K model ("FT-600K"), and (e, f) fine-tuned multi-temperature model ("FT-MultiT"). Parity plots for energies (a, c, e) demonstrate excellent agreement for all models, while force parity plots (b, d, f) show strong overall correlation but subtle differences in error distributions that are not distinguishable visually. These results highlight the necessity of quantitative and property-based benchmarks for robust model evaluation.

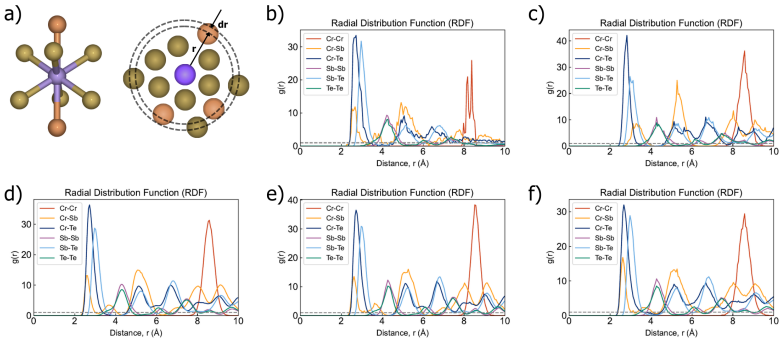



**Fig. S11**: **Radial distribution function (RDF) analysis of Cr-doped $Sb_2Te_3$ at 600 K.** (a) Schematic illustration of the local atomic environment around a Cr dopant (purple) in the $Sb_2Te_3$ lattice, showing the first coordination shell of Te atoms (gold) and second-shell Sb atoms (orange). The dashed circles indicate the radial distances used for RDF calculation. (b-f) Computed RDFs for different atom pairs from molecular dynamics simulations: (b) Ab initio molecular dynamics (AIMD) ground truth reference, (c) MACE foundation model without fine-tuning, (d) MACE model trained from scratch on Cr-$Sb_2Te_3$ data, (e) MACE model fine-tuned with 600 K AIMD data ("FT-600K"), and (f) MACE model fine-tuned with multi-temperature AIMD data ("FT-MultiT"). All simulations were performed at 600 K with 5×5×1 supercells for 200 ps. The RDF peaks correspond to characteristic interatomic distances in the doped structure, with the first peak representing nearest-neighbor correlations.

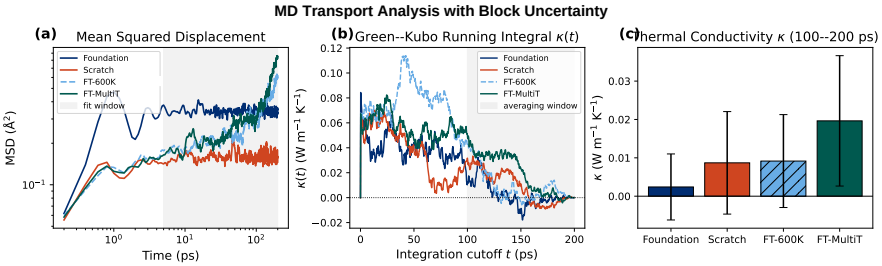



**Fig. S12**: **Transport analysis with block uncertainty.** (a) Mean-squared displacement fits used to estimate diffusion coefficients. (b) Green–Kubo running integrals for thermal conductivity. (c) Block-averaged thermal conductivities over the 100–200 ps integration window. The FT-600K run (dashed/hatched) used a shorter trajectory with a different output interval and is shown for reference.

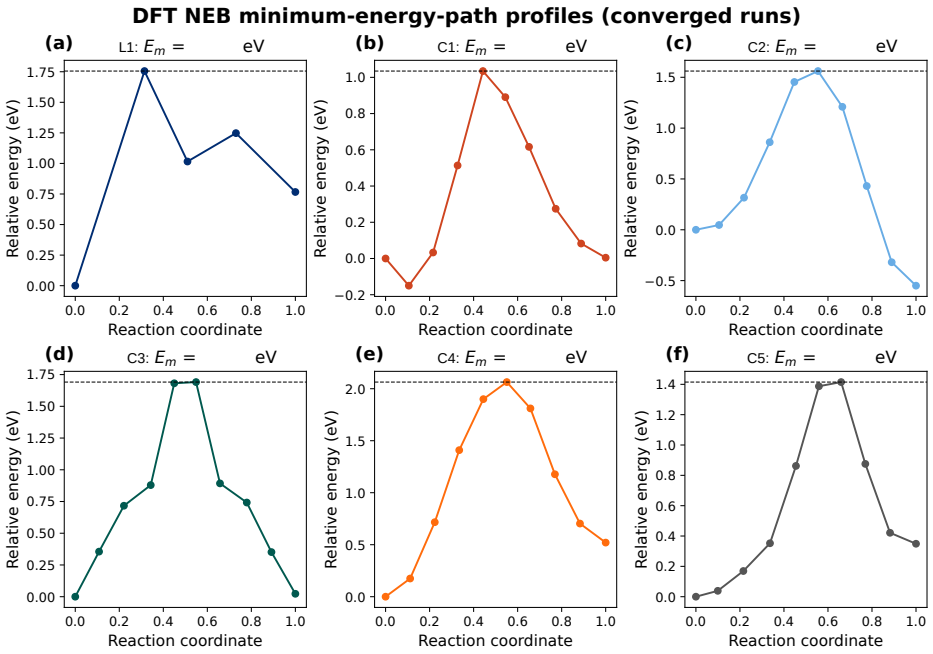



**Fig. S13**: **Audited DFT NEB minimum-energy-path profiles.** Profiles of the within-QL paths L1 and C1–C5; barriers are in Table S6.

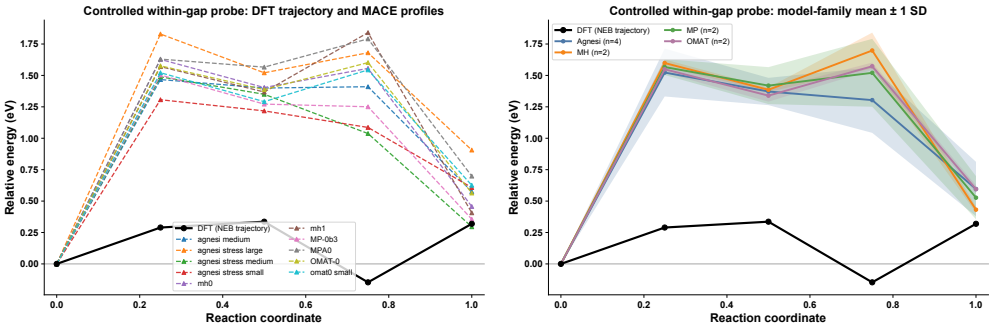



**Fig. S14**: **Foundation-model fixed-path energy profiles for the within-gap path.** (a) Relative-energy profiles for all ten foundation checkpoints evaluated on the same five DFT-path images. (b) Family-aggregated profiles for Agnesi, MH, MP, and OMAT; shaded bands denote pointwise ±1 standard deviation across sub-models within each family, not training-initialization uncertainty. The DFT reference barrier is 0.34 eV. These are fixed-image single-point profiles.

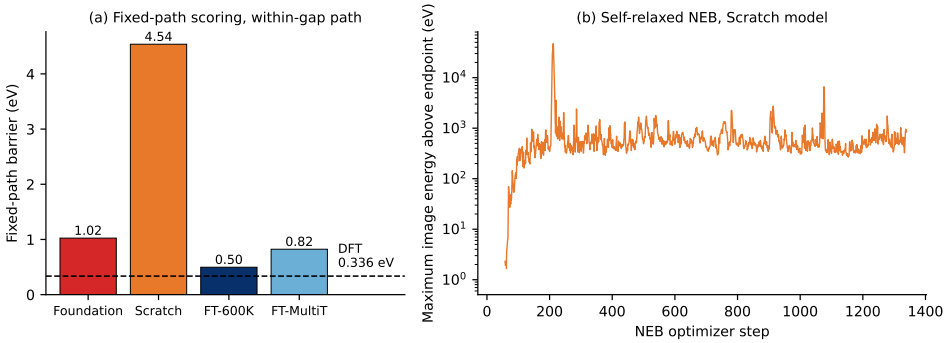



**Fig. S15**: **Fixed-path scoring and self-relaxed NEB.** (a) Barriers assigned by each model to the five DFT images of the 61-atom, one-Cr within-gap path G1; the dashed line is the QE barrier of 0.336 eV. (b) Self-relaxed NEB of the Scratch model on the 82-atom, two-Cr cell: maximum image energy above the fixed endpoint at each optimizer step (logarithmic scale; steps with non-positive values are omitted). The band does not converge.

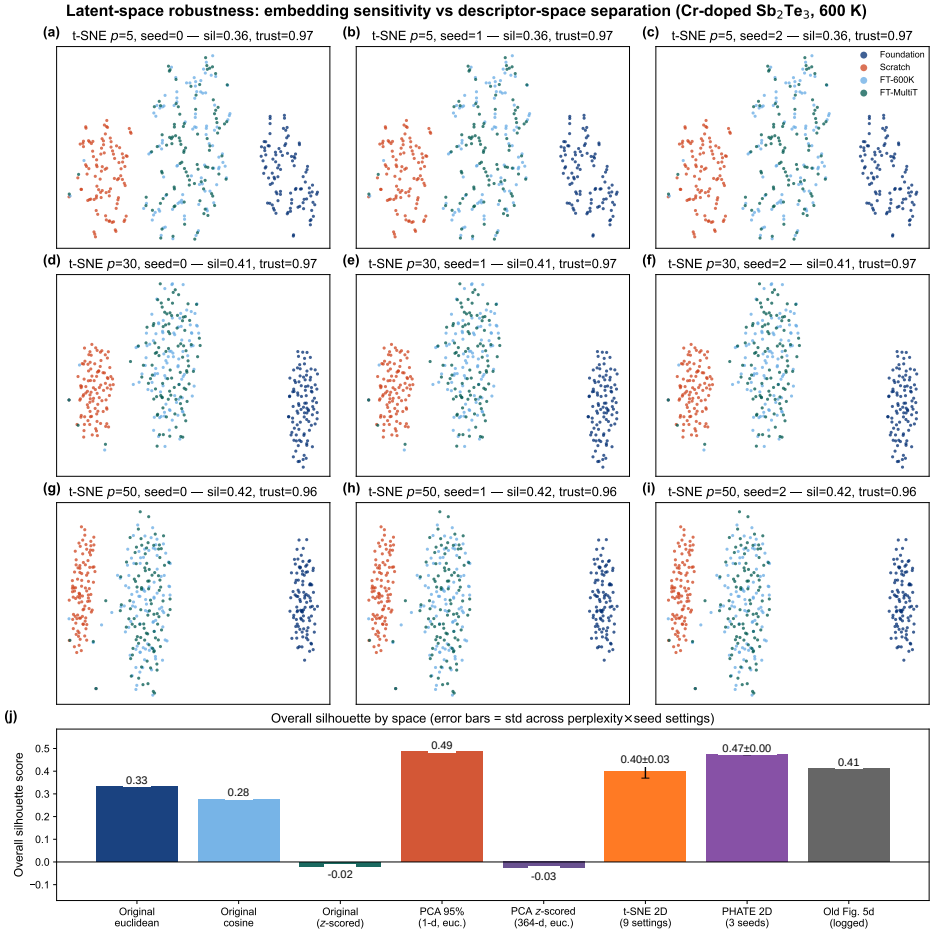



**Fig. S16**: **Latent-space robustness and original-descriptor checks.** The $t$-SNE panels show the audited perplexity range 0.36–0.42 and trustworthiness range 0.963–0.973; nominal seeds at fixed perplexity are identical under the fixed PCA initialization. The PHATE summary is $0.47 \pm 0.00$. The summary panel compares these descriptive projection scores with original-descriptor-space silhouettes of 0.33 (raw Euclidean), 0.28 (cosine), and $-0.02$ (z-scored Euclidean).

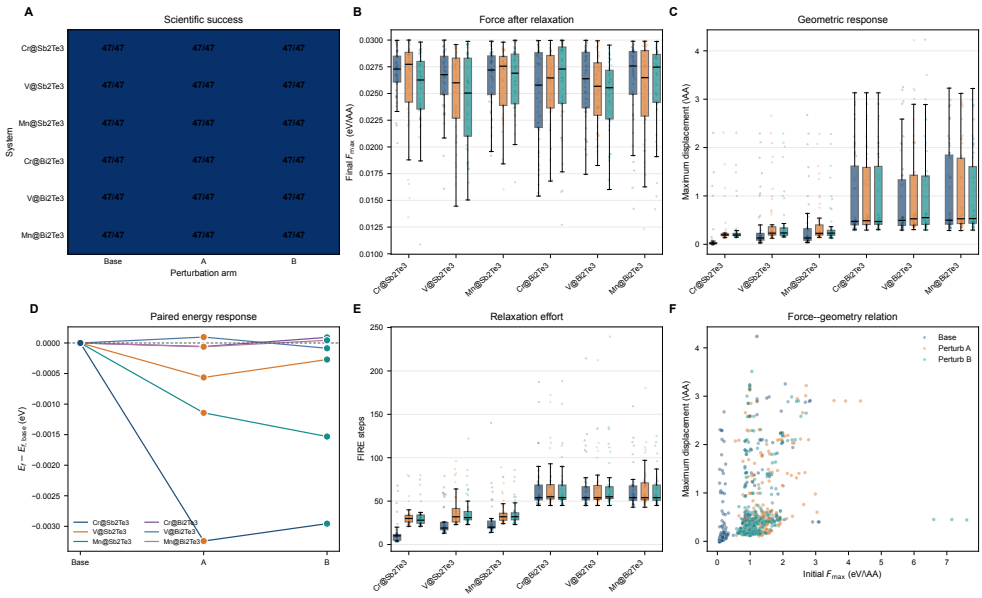



**Fig. S17**: **Campaign overview and relaxation outcomes.** (A) Scientific success across all 846 formal tasks. (B–C) Final-force and maximum-displacement distributions by perturbation arm. (D) Paired energy responses relative to the Base arm. (E) FIRE-step distributions. (F) Initial force versus final displacement. Points and distributions retain endpoint-level variation rather than showing only system medians.

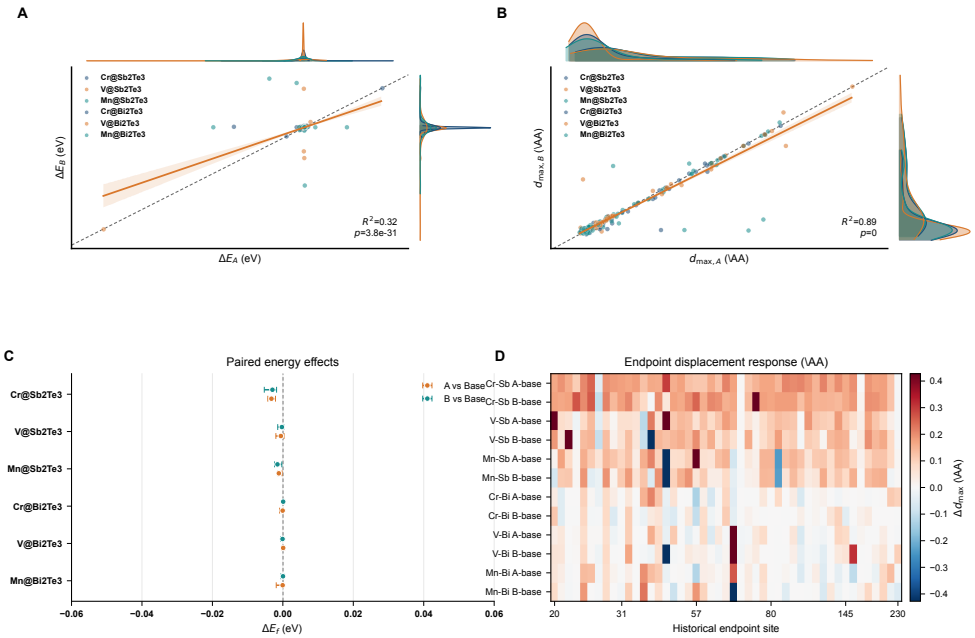



**Fig. S18**: **Perturbation response across systems and endpoints.** Joint distributions compare the energy and geometric response of Perturb A and B. Forest intervals summarize arm-specific changes relative to Base, and the endpoint heatmap exposes the site-level structure hidden by aggregate medians. Intervals are bootstrap summaries of the observed endpoint sample and should not be interpreted as uncertainty in the MACE potential itself.

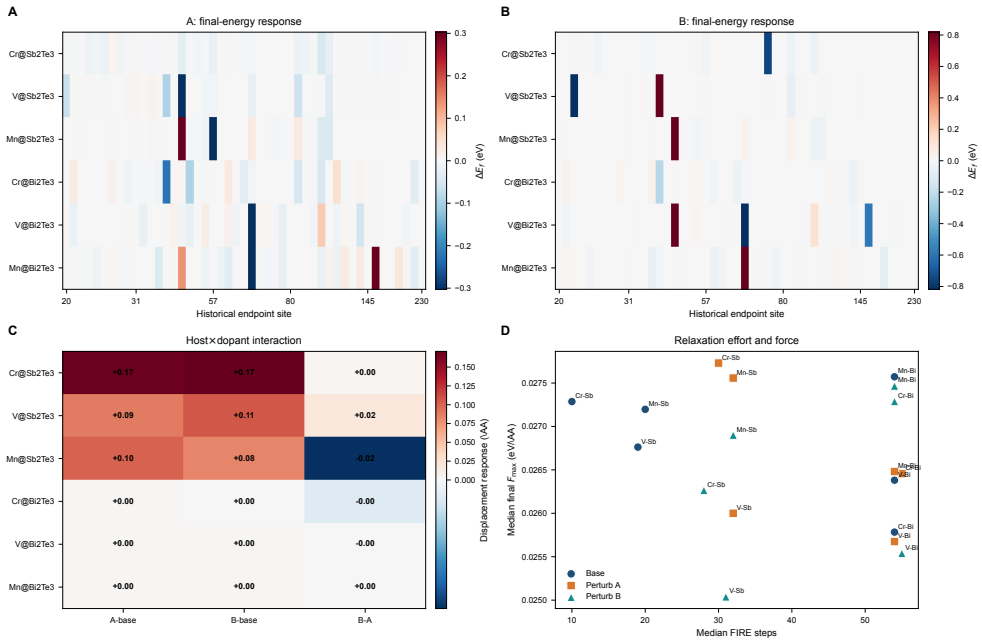



**Fig. S19**: **Host–dopant interactions and relaxation effort.** Endpoint heatmaps show energy response for each host–dopant system, while the interaction map compares the two perturbation arms. The effort panel relates initial force to the number of FIRE steps required by the fixed-cell relaxation protocol.